

Token Usage Tracking

This notebook goes over how to track your token usage for specific calls. It is currently only implemented for the OpenAI API.

Let's first look at an extremely simple example of tracking token usage for a single LLM call.

```
from langchain.llms import OpenAI
from langchain.callbacks import get_openai_callback
```

```
llm = OpenAI(model_name="text-davinci-002", n=2, best_of=2)
```

```
with get_openai_callback() as cb:
    result = llm("Tell me a joke")
    print(cb.total_tokens)
```

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Anything inside the context manager will get tracked. Here's an example of using it to track multiple calls in sequence.

```
with get_openai_callback() as cb:
    result = llm("Tell me a joke")
    result2 = llm("Tell me a joke")
    print(cb.total_tokens)
```

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If a chain or agent with multiple steps in it is used, it will track all those steps.

```
from langchain.agents import load_tools
from langchain.agents import initialize_agent
from langchain.llms import OpenAI
```

```
llm = OpenAI(temperature=0)
```

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```
agent = initialize_agent(tools, llm, agent="zero-shot-react-  
description", verbose=True)
```

```
with get_openai_callback() as cb:  
    response = agent.run("Who is Olivia Wilde's boyfriend? What is his  
    current age raised to the 0.23 power?")  
    print(cb.total_tokens)
```

> Entering new AgentExecutor chain...

I need to find out who Olivia Wilde's boyfriend is and then calculate his age raised to the 0.23 power.

Action: Search

Action Input: "Olivia Wilde boyfriend"

Observation: Jason Sudeikis

Thought: I need to find out Jason Sudeikis' age

Action: Search

Action Input: "Jason Sudeikis age"

Observation: 47 years

Thought: I need to calculate 47 raised to the 0.23 power

Action: Calculator

Action Input: $47^{0.23}$

Observation: Answer: 2.4242784855673896

Thought: I now know the final answer

Final Answer: Jason Sudeikis, Olivia Wilde's boyfriend, is 47 years old and his age raised to the 0.23 power is 2.4242784855673896.

> Finished chain.

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